

Cendong Lu

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EDUCATION

University of Auckland Master of Information Technology		Auckland, NZ 2025.11-2027.1
Purdue University Master of Mechanical Engineering	GPA 3.7	Indiana, U.S 2017.1-2019.5
North University of China Bachelor of Thermal energy and power engineering		Taiyuan, China 2012.8-2016.8

WORK EXPERIENCE

Shanghai Marvel Tech Co.Ltd Turbomachinery design engineer	2020/08/15 – 2021/11/14
• Join the TRT blade optimization project through STAR CCM+ • Join the secondary air system design for 4MW hydrogen gas turbine through STAR CCM+ • Join the one-dimension calculation units' development for secondary air system by MATLAB • Join Airplane APU inlet duct optimization process development by HEEDS	
Noveastern Co.Ltd CAE Leader	2021/11/15 – 2023/06/30
• Handle with Vehicle Air Intake System CFD analysis and report writing. • Handle with e-Vehicle battery cooling base CFD analysis and report writing. • Build a UDF java code in Star CCM+ to accomplish auto running and report export for Vehicle Air Intake System CFD analysis	
Alten China Battery Pack Thermal Simulation Engineer	2023/10/30 – 2024/03/15
• Participate in CFD pressure drop and HTC analysis of cooling plate for the battery pack of the company's electrical vehicle model.	

AI Application Developer

Key Projects:

Testcase Generation Helper

Software/Hardware Built automated testing workflows based on requirements, standards documentation, and historical test cases to assist testing teams with new requirements.

-Technologies: Dify, RAG, Azure DevOps, SQL, Docker

AI Knowledge-Base Assistant for Testcase Generation Helper (RAG-boosted)

Built a full-stack knowledge base to support RAG for Testcase Generation Helper.

Storing historical test cases to aid future test-case AI-generation.

React front end: login, department Knowledge Base Data upload, Knowledge Base Data management.

Python RESTful API: secure auth, file upload/delete/download, job queue.

UI Automation Agent

Built a UI-agent that turns plain-language task lists into hands-free runs of internal Windows apps for the QA team.

Core flow: multi-modal perception (VLM + UIA) → LLM action planner → pywinauto executor → execution-result verifier → self-correction loop.

Stack: AI agent, Windows UIA, pywinauto, VLM, LLM.

Training Experience**AI Job-Oriented Training**

2024/04 – 2024/10

- Studied Python basic and advanced grammar.
- Studied Linux system operation commands, MySQL database usage, and Pandas data processing and analysis
- Studied machine learning algorithms
- Practice the application of deep learning framework PyTorch and deep learning core models - convolutional neural networks; recurrent neural networks.
- Studied natural language processing (NLP) related knowledge.
 - Text data processing.
 - HMM and CRF models.
 - Attention mechanism, encoder, decoder and Transformer architecture.
 - NLP transfer learning and the use of pre-trained model BERT.

SKILLS

• AI & Automation Development

- Cursor, Jupyter Notebook, low code workflow tool Dify, Agno, LangGraph
- Expert in Python programming with NumPy, Pandas for automated data processing and API integrations.
- Advanced knowledge of MySQL, Neo4j graph databases, Redis for automated data management systems.
- Experienced with machine learning algorithms: KNN, LR, CART, K-means, RF, GBDT, XGBoost for intelligent automation.
- Proficient in PyTorch, Attention mechanisms, Transformer, BERT, CNN, LSTM, GRU, Seq2Seq for AI-powered solutions.
- Specialized in NLP automation: word vector training, text generation, tokenization, sequence labeling, named entity recognition, keyword extraction.

• Certifications

- AWS Certified Cloud Practitioner

Dec 2025 - Dec 2028

<https://www.credly.com/go/EzfvAGUMJAYhAjVwn9NWCQ>

- AWS Certified AI Practitioner

Jan 2026 - Jan 2029

<https://www.credly.com/badges/eff6eb78-da32-4ea9-8d8c-adeef73a68ff/seek>